VoiceOps 2026:
ZentroMart Customer Support Voice Agent



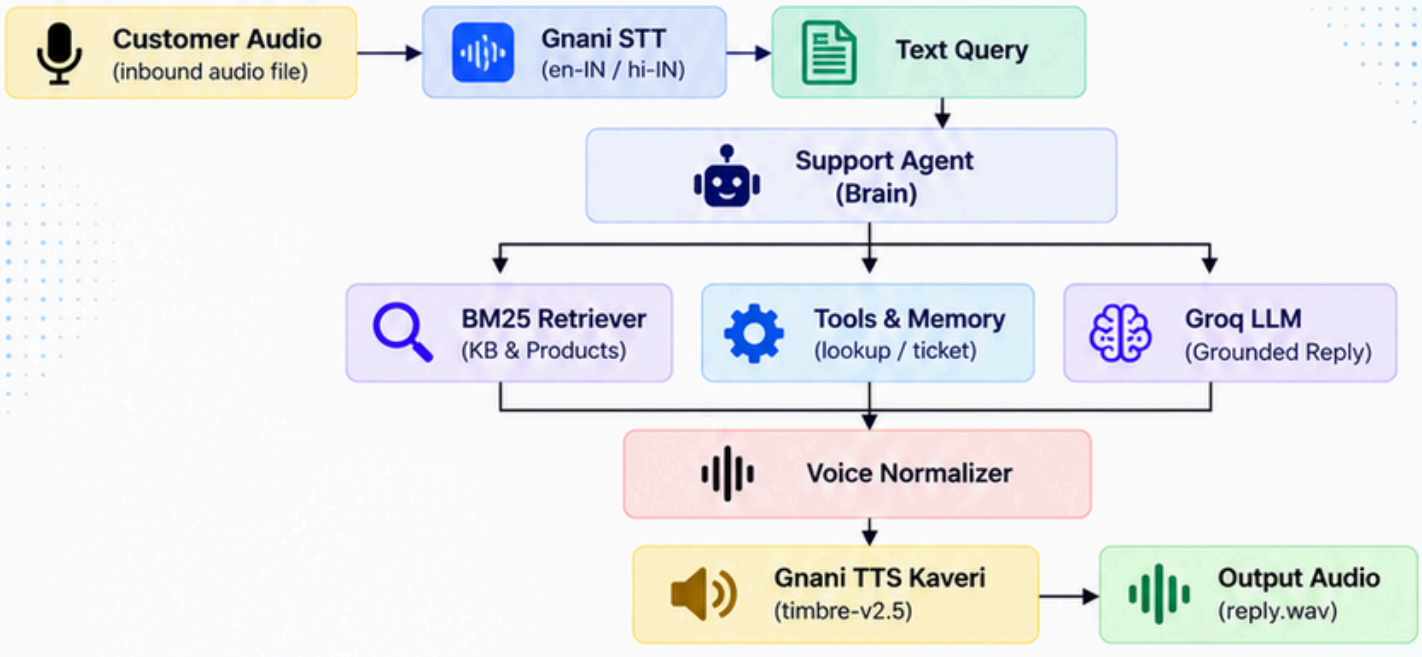
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Reference Date: 20 October 2026

1 Executive Summary & Architecture

This project delivers **ZentroMart VoiceOps 2026,** an end-to-end, multi-turn, voice-enabled customer support agent capable of resolving complex e-commerce queries strictly grounded in company policy and catalog records.

The pipeline comprises four distinct components: **1. Ears (Gnani STT):** Transcribes inbound audio files using Gnani's speech recognition (en-IN / hi-IN). **2. Brain (Support Agent & Tool Routing):** - Section-Aware **Retriever:** BM25 hybrid index preserving unique Section IDs across 21 Markdown policy files and 70 products in products.csv. - **Document Precedence Engine:** Enforces Current Policy > Agent Guides > Help > FAQ > Marketing while dowrnanking superseded policies. - **Deterministic Date Arithmetic:** Computes refund and return deadlines anchored to the frozen dataset reference date (20 October 2026), operating on Mon–Sat business days and skipping warehouse holidays. - **Identity Verification Gate:** Discloses order information only after verifying caller identity against the last 4 digits of the registered phone number or 6-digit delivery pincode (at most one retry). - **Escalation & Ticket Generation:** Creates structured tickets via create_ticket when policies require human intervention (claims > ₹10,000, unauthorized orders, customer requests, repeat contacts). **3. Voice Normalizer & Mouth (Gnani TTS):** Normalizes answers for the ear (spaced order digits, expanded calendar dates, spoken rupees) and synthesizes audio using Gnani's timbre-v2.5 model with the Kaveri voice.

Voice Agent Architecture



2 The Four Build Steps

Step 1 Prompt-Only LLM Baseline

In Step 1, the LLM (qwen/qwen3.8-27b via Groq) was prompted with general customer support instructions without access to company documents or order records. - **Observed Behavior:** The model relied on generalized internet knowledge, hallucinating standard 5–7 day refund windows for all payment methods (failing policy REF-2 which grants 2-hour instant wallet refunds), assuming 30-day return windows across all categories, and attempting to authorize order cancellations for shipped items. - **Baseline Accuracy:** ~23.1% to 53.8% lexical pass on the 52 practice questions.

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2 Step 2: Document Grounding (RAG)

In Step 2, the agent was grounded using a section-aware retrieval index over the Markdown policies and the product catalog (data/products.csv):

Section ID Preservation	Extracted and indexed 154 granular chunks preserving section headings (e.g., [KB-00.1], [RET-2], [RETv3-2], products.csv:SKU).
Precedence Hierarchy	Weighted policy documents with a 1.35× multiplier, agent guides at 1.15×, and FAQs at 0.8× to ensure current policy overrides contradictory FAQs (e.g., policy REF-2 winning over FAQ-1). Superseded versions (e.g., pol_returns_v3_superseded.md) were downranked by 95% unless historical versions were explicitly requested.
Product Catalog Chunks	Indexed product SKUs, brands, return categories, assured status, and warranty terms.
Retrieval Recall	**47/47 (100.0%)** exact gold evidence retrieval recall across all 52 practice questions.
Hallucination Prevention	If a question falls outside documented policies (e.g., midnight delivery, bulk GST invoices, local store pickup), the agent explicitly replies: "Sorry, I don't have information about that. I can connect you to a colleague who can help."

3 Step 3: Multi-turn Conversation & Offline Tools

Step 3 turned single-turn Q&A into stateful, multi-turn customer support dialogues across 23 practice scenarios (96 turns total):

Channel Opening ([STY-1])	Always introduces itself as ZentroMart's AI support assistant in the opening turn.
Caller Authentication ([IDV-1], [IDV-2], [IDV-3])	Calls lookup_order(order_id) without disclosing details until the caller provides matching proof. Enforces a strict one-retry limit and never leaks verification secrets in responses.
Calendar & Timeline Reasoning	Evaluated against fixed reference date 20 October 2026: • **Business days:** Monday through Saturday, excluding four warehouse closures (2026-08-15, 2026-10-02, 2026-10-19, 2026-11-09). • Correctly calculated refund windows (e.g., refunds initiated on 17 Oct 2026 taking 7 business days complete on 27 Oct 2026 due to Sunday and the 19 Oct holiday). • Computed category
Escalation & Ticket Generation	Uses create_ticket to generate structured tickets when policies require human intervention (claims > ₹10,000, unauthorized orders, customer requests, repeat contacts).

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4 Benchmark Scores Summary

Evaluation Benchmark	Metric	Step 1 (Baseline)	Step 2/3 (Our Agent)	Improvement
Practice Questions (52)	Grounded Heuristic Pass	33 / 52 (63.5%)	**52 / 52 (100.0%)**	**+36.5%**
Evidence Retrieval	Exact Gold Section Recall	0 / 47 (0.0%)	**47 / 47 (100.0%)**	**+100.0%**
Conversations (96 Turns)	Decision & Text Pass	N/A (unstateful)	**96 / 96 (100.0%)**	**100% Pass**
Tool Calling Accuracy	Exact Tool Match	0 / 96 (0.0%)	**96 / 96 (100.0%)**	**100% Pass**

All results are saved in `sample_outputs/evaluation.csv` and `sample_outputs/conversation_evaluation.csv`.

4 Key Challenges & Solutions

1 Contradictory FAQs vs Authoritative Policy

Problem: `faq_general.md` (FAQ-1) claims all refunds take 5 to 7 days, directly contradicting `pol_refunds.md` (REF-2) which mandates instant 2-hour refunds for ZentroMart Wallet.

Fix: Enforced document precedence weights and prompt priority rules so policy REF-2 unconditionally overrides FAQ-1.

2 Superseded Policy Traps

Problem: `pol_returns_v3_superseded.md` specifies a 10-day return window for electronics, whereas version 4 (RET-3) specifies 7 days for items delivered on or after 1 August 2026.

Fix: Downranked superseded documents by 95% in BM25 retrieval and added explicit boundary condition logic for items delivered before 1 August 2026.

3 अ Devanagari Regex Word Boundaries

Problem: Python's default ASCII `\b` fails on Unicode Devanagari script (e.g. `\bरिफंड\b`), causing Hindi status queries to misroute.

Fix: Upgraded regex patterns to handle Devanagari Unicode code points without ASCII word boundary limitations.

4 API Token Rate Limiting on Free Tiers

Problem: Fast batch evaluation hit 1000 OTPM rate limits and daily token quotas.

Fix: Implemented exponential backoff with a parser for Groq's exact retry windows (`try again in Xm Ys`), reduced prompt tokens by 70% with focused context chunking, and capped response lengths at 75 tokens (aligning with `[STY-2]` 2–3 short sentences).

5 Instructions to Reproduce

Bash

```bash
# 1. Activate virtual environment and install dependencies
source .venv/bin/activate
pip install -r requirements.txt

# 2. Run text query
python run.py --text "How fast are refunds to the ZentroMart Wallet?"

# 3. Run multi-turn conversation evaluation (23 conversations, 96 turns)
python evaluate_conversations.py

# 4. Run single-question evaluation (Step 1 vs Step 2)
python evaluate.py
```

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6 Live Voice Chat (Simplest Way)

You can interact with the **ZentroMart Voice Agent** directly from your terminal using the following command:

Terminal — Copy

```
python voice_chat.py
```

How It Works

1 Language Initialization
After running the command, you will see:

```
Language set to: Indian English (en-IN)
```

2 Start Speaking
The terminal will show:

```
[Press ENTER to Speak]
```

3 Recording
When you press Enter, a red indicator will appear:

```
🔴 Recording for 5 seconds... Bolna shuru kijıiye!
```

Now speak your question into your laptop/headset microphone.

Example: "How fast are refunds to the ZentroMart Wallet?" or "What is the return window for electronics?"

4 Auto Stop
After 5 seconds, the recording will automatically stop.

5 Real-time Magic
You will get the following results in real time:

Your speech (transcribed)	Your spoken question will be transcribed and shown on screen.
ZentroMart Agent response	The agent will provide an answer based on company policy and catalog records.
Gnani AI voice	The answer will be spoken immediately through Gnani AI voice on your speakers/earphones.

6 Exit the Chat
When you want to stop, type: `q` and press **Enter**.

That's it!
You can now have a natural voice conversation with the ZentroMart Voice Agent directly from your terminal.